Rank inequalities force orbit degeneration for 2×2 tuples

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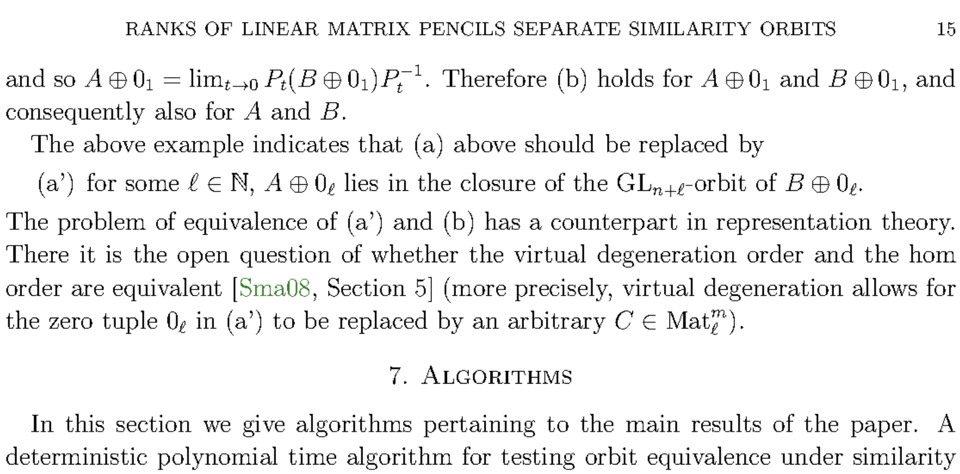
Status. Partial result, likely valid, pending human review.

1 Source question

Derksen, Klep, Makam, and Volčič [1] ask at the end of Section 6 whether their rank inequalities are equivalent to stable orbit degeneration. In their notation, for $A, B \in \mathrm{Mat}_n(k)^m$, condition (b) is

$$\operatorname{rank}\left(I \otimes T_0 + \sum_i A_i \otimes T_i\right) \leq \operatorname{rank}\left(I \otimes T_0 + \sum_i B_i \otimes T_i\right) \tag{R}$$

for all square affine linear matrix pencils, while (a') says that for some ℓ, $A \oplus 0_\ell$ lies in the closure of the similarity orbit of $B \oplus 0_\ell$. This is related to the open comparison between the hom order and virtual degeneration order.


RANKS OF LINEAR MATRIX PENCILS SEPARATE SIMILARITY ORBITS 15

and so $A \oplus 0_1 = \lim_{t\to 0} P_t(B \oplus 0_1)P_t^{-1}$. Therefore (b) holds for $A \oplus 0_1$ and $B \oplus 0_1$, and consequently also for A and B.

The above example indicates that (a) above should be replaced by

(a') for some $\ell \in \mathbb{N}$, $A \oplus 0_\ell$ lies in the closure of the $\mathrm{GL}_{n+\ell}$-orbit of $B \oplus 0_\ell$.

The problem of equivalence of (a') and (b) has a counterpart in representation theory. There it is the open question of whether the virtual degeneration order and the hom order are equivalent [Sma08, Section 5] (more precisely, virtual degeneration allows for the zero tuple 0_ℓ in (a') to be replaced by an arbitrary $C \in \mathrm{Mat}_\ell^m$).

7. ALGORITHMS

In this section we give algorithms pertaining to the main results of the paper. A deterministic polynomial time algorithm for testing orbit equivalence under similarity

2 Idea of the proof

The source's matricization identifies (R) with inequalities $\dim \mathrm{Hom}(X, M_A) \geq \dim \mathrm{Hom}(X, M_B)$ for all finite-dimensional modules X. In dimension two there is almost no room. A nonsemisimple module is a nonsplit extension of two one-dimensional simples and has a unique proper nonzero submodule. Testing first with that submodule and then with the whole extension forces M_A either to be the same extension or to contain both simple factors as submodules. The latter means that M_A is the split extension, which is visibly a limit of the nonsplit one. Thus stabilization is unnecessary in this dimension. The general question for $n \geq 3$ remains open.

3 Result and proof

Let k be algebraically closed and let an m-tuple act on k^n as a finite-dimensional module over the free algebra $k\langle x_1, \ldots, x_m \rangle$.

Theorem 1. *For $A, B \in \mathrm{Mat}_2(k)^m$, condition* (R) *holds if and only if $A \in \overline{B^{\mathrm{GL}_2}}$. Consequently (a') and (b) in the source are equivalent for $n = 2$, and one may take $\ell = 0$.*

Proof. Orbit degeneration implies (R) by lower semicontinuity of matrix rank. For the converse, let $M = M_A$ and $N = M_B$. Lemma 3.1 and the padding argument in [1] translate (R) into

$$\dim \mathrm{Hom}(X, M) \geq \dim \mathrm{Hom}(X, N) \tag{H}$$

for every finite-dimensional module X.

Suppose first that N is semisimple. If N is simple, (H) with $X = N$ gives a nonzero map $N \to M$; it is injective and, dimensions being equal, an isomorphism. If $N = S \oplus T$ for distinct one-dimensional simples, (H) with $X = S, T$ puts a copy of each in the socle of M, so $M \cong S \oplus T$. If $N = S \oplus S$, then $\dim \mathrm{Hom}(S, M) \geq 2$; the common S-eigenspace is all of M, again giving $M \cong N$.

It remains that N is a nonsplit extension

$$0 \longrightarrow S \longrightarrow N \longrightarrow T \longrightarrow 0 \tag{E}$$

of one-dimensional simples. It has a unique proper nonzero submodule, namely S. Moreover, $\dim \mathrm{Hom}(S, N) = 1$, so (H) puts a copy of S in M. Moreover,

$$\dim \mathrm{Hom}(N, M) \geq \dim \mathrm{End}(N).$$

If one of these maps $N \to M$ has rank two, then $M \cong N$.

Assume otherwise that every such map has rank at most one. When $S \ncong T$, the endomorphism algebra of the nonsplit extension is one-dimensional, so there is a nonzero rank-one map $N \to M$. Its kernel must be the unique submodule S of N, and hence its image is a copy of T. Thus M contains the two distinct simple submodules S and T, and $M \cong S \oplus T$.

When $S \cong T$, write, in a suitable basis, $B_i = \alpha_i I + b_i E_{12}$, with some $b_i \neq 0$. Then $\mathrm{End}(N) = \mathrm{span}\{I, E_{12}\}$ has dimension two. Every rank-one map $N \to M$ kills the unique submodule S and factors through the fixed quotient $N \twoheadrightarrow S$. Hence (H) forces $\dim \mathrm{Hom}(S, M) \geq 2$. The common S-eigenspace is therefore all of M, so $M \cong S \oplus S$.

In either nonisomorphic case $M \cong S \oplus T$. Finally, writing $B_i = \left(\begin{smallmatrix} \alpha_i & b_i \\ 0 & \beta_i \end{smallmatrix}\right)$, conjugation by $\mathrm{diag}(t, 1)$ multiplies every b_i by t. Letting $t \to 0$ shows that $S \oplus T$ lies in $\overline{N^{\mathrm{GL}_2}}$. Thus $A \in \overline{B^{\mathrm{GL}_2}}$ in every case. □

4 Verification, limitations, and novelty

The proof uses no computation. Its cases exhaust all two-dimensional modules over an algebraically closed field: simple, semisimple of length two, or a nonsplit length-two extension. The theorem does not address non-algebraically closed fields or $n \geq 3$. A bounded search of the local corpus and exact web phrases "virtual degeneration order" and "hom order" through 22 July 2026 found no later solution of the general question. No claim is made that this elementary low-dimensional subcase is absent from folklore.

Human-review recommendation. Check especially that the source's rectangular Hom matricization is padded without changing the direction of (R), and the dimension-two self-extension argument.

References

[1] H. Derksen, I. Klep, V. Makam, J. Volčič, *Ranks of linear matrix pencils separate simultaneous similarity orbits*, arXiv:2109.09418, especially Lemma 3.1 and Section 6.